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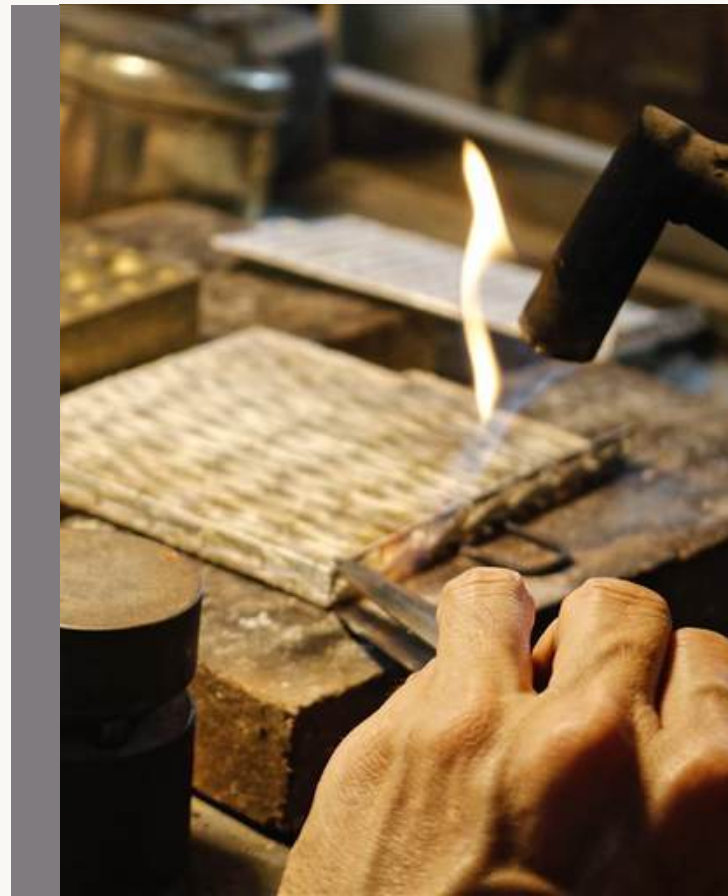


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We are Group 2 of the Marketing Data Analysis course at the International School, Vietnam National University, and we would like to extend our sincere appreciation to all individuals who supported and contributed to the completion of this final report entitled “Determinants of Attitudinal Loyalty in E-commerce Platforms for Traditional Vietnamese Artisanal Products: A Case Study of Đậu Bạc.”

First and foremost, we would like to express our deepest gratitude to Ph.D. Bui My Trinh for her continuous guidance, academic insight, and constructive feedback throughout the research process. Her professional expertise, thoughtful supervision, and encouraging support played a vital role in shaping the direction, analytical rigor, and overall quality of this study.

We also sincerely thank all respondents who participated in the survey and provided valuable data for the analysis. Their cooperation and honest responses formed the empirical foundation of this research and made the application of marketing data analytics possible.

Finally, we would like to acknowledge the dedication, collaboration, and mutual support among the members of Group 2. Through this project, we have not only strengthened our analytical and research skills but also developed effective teamwork and professional communication. This report represents the collective effort of the group and reflects both our academic growth and shared commitment throughout the course.

We are truly grateful to everyone who contributed to making this project a meaningful and rewarding learning experience.

Abstract

The expansion of e-commerce has fundamentally transformed consumer interactions with traditional artisanal products, creating new challenges in sustaining customer loyalty in digital environments. In Vietnam, traditional silver jewelry such as *Đậu Bạc* faces increasing pressure to adapt to online platforms where experiential and cultural values must be conveyed without physical contact. This study examines how digital shopping experiences influence attitudinal loyalty toward traditional silver jewelry on Vietnamese e-commerce platforms.

The research aims to identify key platform-related and psychological determinants shaping consumers' attitudinal loyalty and to explore heterogeneity in online shopping experiences. Drawing on the Online Customer Shopping Experience (OCSE) framework, the study analyzes eight constructs, including interactivity, informativeness, visual engagement, navigation ease, trust, convenience, enjoyment, and self-control, with attitudinal loyalty as the focal outcome variable.

The study adopts a quantitative research design using a standardized survey dataset provided by the course instructor. A structured data analytics pipeline was employed, including descriptive statistics, reliability analysis (Cronbach's Alpha), exploratory factor analysis (EFA), K-means cluster analysis, correlation analysis, group comparison tests (independent samples t-test and one-way ANOVA), and multiple linear regression. These analytical techniques allow for both measurement validation and empirical examination of relationships among key constructs.

The results indicate that experiential and psychological dimensions play a central role in shaping attitudinal loyalty, while demographic factors exhibit limited explanatory power. Overall, the study demonstrates how marketing data analytics can be used to systematically analyze online consumer behavior and generate insights relevant to traditional artisanal products operating in e-commerce contexts.



INTRODUCTION

The rapid expansion of e-commerce has transformed marketing from transactional exchanges into relationship-oriented interactions. Prior studies suggest that brand–consumer relationships function similarly to interpersonal relationships, in which trust and emotional attachment play a decisive role in sustaining loyalty (Aaker, Fournier, & Brasel, 2004). When such relationships are weakened, customer disengagement and declining loyalty are likely to occur.

Within this relational framework, brand trust serves as a key mechanism linking customer perceptions to loyalty outcomes. Chaudhuri and Holbrook (2001) show that trust enhances both purchase likelihood and attitudinal loyalty, particularly in digital environments where physical product inspection is absent and perceived risk is higher.

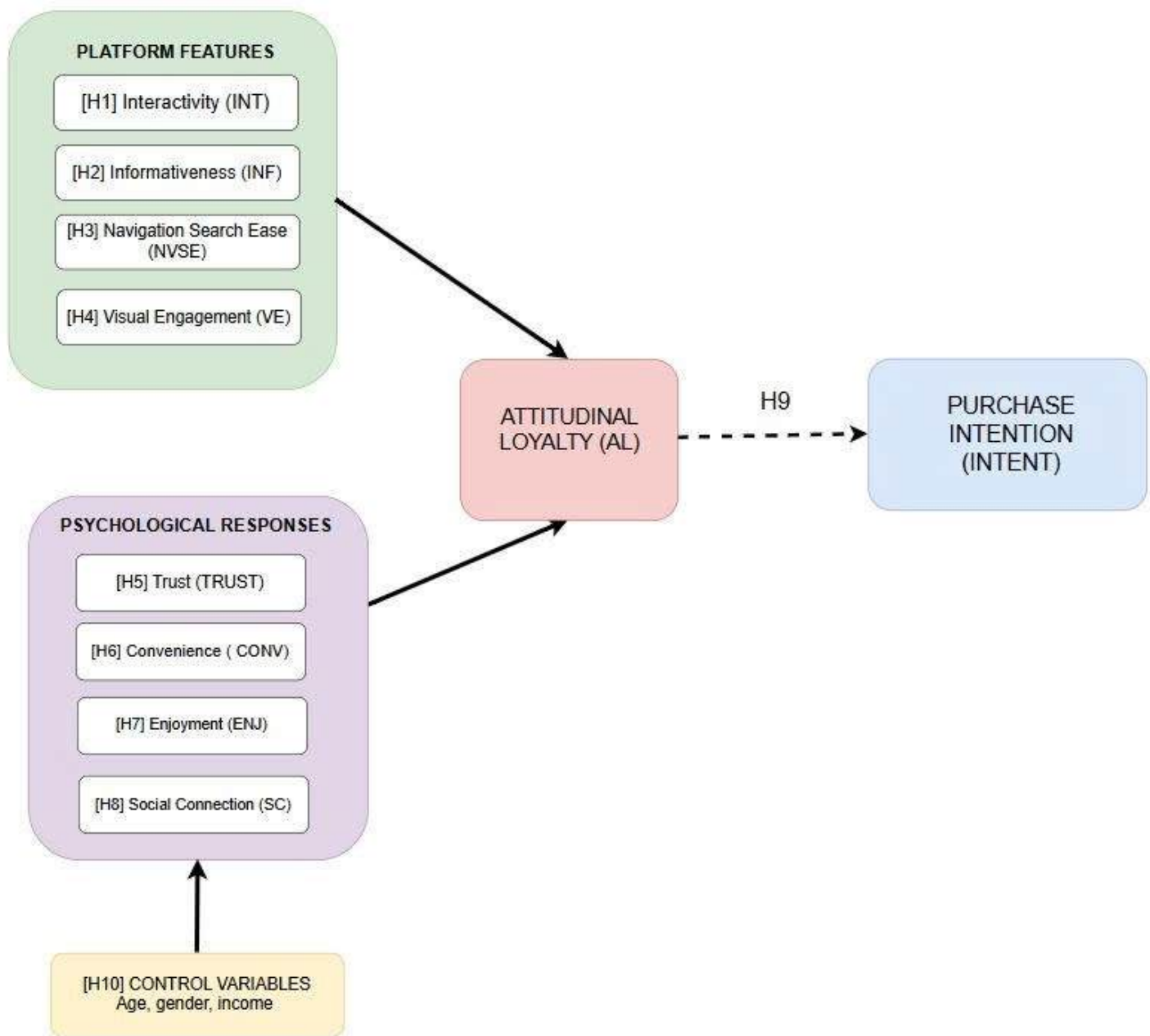
Research on Online Customer Shopping Experience (OCSE) further conceptualizes online shopping as a holistic process that encompasses cognitive, emotional, and behavioral responses arising from interactions with digital platforms (Rose et al., 2012). This perspective emphasizes that online experiences are shaped by both functional system features and psychological evaluations.

For traditional artisanal products such as Đậu Bạc, these issues are especially salient. While e-commerce enables wider market access, the lack of physical interaction makes it difficult to convey craftsmanship and cultural authenticity. Consequently, e-commerce platforms function not only as sales channels but also as experiential intermediaries that mediate cultural value.

Accordingly, platform features and psychological responses—including informativeness, visual engagement, interactivity, trust, convenience, enjoyment, and self-control—are expected to play a critical role in shaping online shopping intention and attitudinal loyalty toward traditional Vietnamese products. Examining these relationships is therefore essential for understanding how e-commerce can support the sustainable development of heritage-based brands in Vietnam’s digital economy.



Conceptual Framework



Conceptual Model

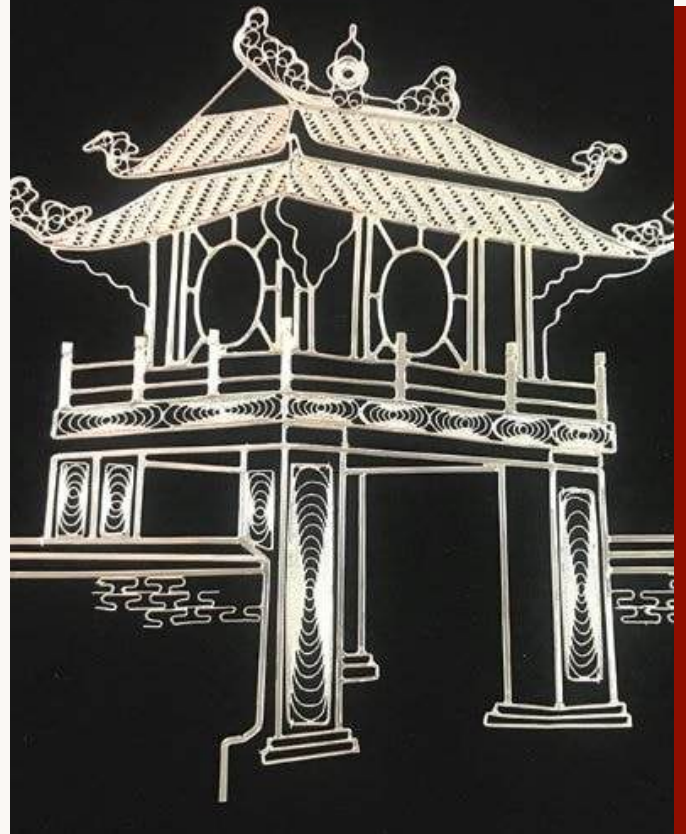
Conceptual Framework

01 Interactivity (INT)
The ability for users to communicate, exchange comments, and receive advice from others on the platform.

02 Informativeness (INF)
The extent to which the platform provides complete, detailed, and truthful information about the products.

03 Visual Engagement (VE)
The use of smart, intuitive designs, high-quality images, and appealing aesthetics to satisfy the user's visual experience.

04 Navigation Ease (NVSE)
The simplicity of searching for and navigating through similar or recommended products on the platform.



05 Trust (TRUST)
The user's confidence in the platform's reputation and the feeling of security during transactions.

06 Convenience (CONV)
The ability to shop anytime and anywhere, saving the consumer significant time and effort.

07 Enjoyment (ENJ)
The feelings of happiness and excitement derived from the shopping process and participating in platform programs.

08 Self-Control (SC)
The consumer's tendency to evaluate long-term product value and reconsider purchases rather than buying impulsively.





Hypothesized Relationships

Interactivity has a positive impact on consumers' Attitudinal Loyalty. H1

Informativeness positively influences Attitudinal Loyalty toward artisanal products. H2

Visual Engagement significantly affects Attitudinal Loyalty. H3

Navigation Ease is positively related to Attitudinal Loyalty. H4

Trust acts as a key driver for customer loyalty on e-commerce platforms. H5

Convenience positively impacts Attitudinal Loyalty for local cultural goods. H6

Enjoyment during the online shopping process enhances Attitudinal Loyalty. H7

Self-Control positively influences Attitudinal Loyalty toward traditional artisanal products. H8

Platform features collectively have a positive impact on the user's Online Shopping Intention. H9

Online Shopping Intention has a positive and significant effect on Attitudinal Loyalty. H10

Market Gap & Analytical Motivation

Despite the extensive body of research on e-commerce, this study identifies three critical gaps that remain insufficiently addressed in the existing literature.

First, the lack of integration between functional and psychological dimensions of online shopping experience. Most prior studies on online consumer shopping experience (OCSE) have predominantly focused on functional aspects such as usability, efficiency, or ease of navigation. However, as noted by Gulfraz et al. (2022), there is limited empirical evidence that provides a comprehensive understanding of how psychological responses such as self-control and enjoyment interact with platform features to shape long-term consumer loyalty. This study addresses this gap by integrating four system-related platform features and four psychological responses into a unified analytical framework for traditional artisanal products, specifically *Đậu Bạc* jewelry.

Second, ambiguity in the measurement and conceptualization of website design and user interaction. Garrett et al. (2016) emphasize that there is still no clear consensus on how website aesthetics and usability should be operationalized and evaluated in predicting user engagement and loyalty. In the context of traditional handicrafts, where visual refinement and aesthetic presentation are particularly important, existing studies have not sufficiently clarified which specific design-related factors such as visual engagement effectively foster attitudinal loyalty toward local cultural products in Vietnam's e-commerce environment.

Third, the tension between the "sincerity" of heritage-based products and the "intangible" nature of digital shopping environments. According to Ran and Wan (2023), the effectiveness of branding strategies is highly contingent on brand personality, particularly the distinction between sincerity and excitement. Traditional craft products such as *Đậu Bạc* inherently embody a "sincere" brand personality rooted in authenticity, heritage, and craftsmanship. In contrast, e-commerce platforms often emphasize speed, convenience, and stimulation, which may encourage impulsive purchasing behaviors. To date, little research has examined how digital platforms can sustain attitudinal loyalty toward heritage-based brands while preserving consumers' sense of autonomy when engaging in online shopping experiences.

To address these gaps, this study applies marketing data analysis to quantify the effects of eight key determinants four platform features (interactivity, informativeness, visual engagement, and navigation ease) and four psychological responses (trust, convenience, enjoyment, and self-control) on consumer loyalty toward *Đậu Bạc* products in the Vietnamese e-commerce context.





ĐỘC ĐÁO NGHỀ ĐẬU BẠC ĐỊNH CÔNG

RESEARCH OBJECTIVES

First, to apply Factor Analysis to identify and validate the underlying dimensions of the online shopping experience, ensuring the measurement quality of the 8 key constructs: interactivity, informativeness, visual engagement, navigation ease, trust, convenience, enjoyment, and self-control.

Third, to identify and rank the factors that most strongly influence online shopping intention (i.e., the intention to continue shopping online, coded as Yes/No). This will be addressed using correlation analysis and logistic regression to estimate the effect size and statistical significance of each determinant.

Finally, to provide actionable intelligence for traditional craft-makers to enhance their digital presence, balancing functional efficiency with the "sincere" storytelling of their heritage.

The primary aim of this study is to investigate the factors driving consumer loyalty and purchase behavior toward traditional artisanal products (specifically Đậu Bạc jewelry) on Vietnamese e-commerce platforms. To achieve this, the study identifies several specific objectives:

Second, to segment customers into distinct groups using Cluster Analysis based on their perceptions of platform features and psychological responses, allowing for personalized marketing strategies for Đậu Bạc.

Fourth, to examine the role of Attitudinal Loyalty as a core outcome. The study evaluates how platform features and psychological responses influence loyalty toward local cultural goods and whether this loyalty acts as a significant predictor for future shopping intentions.

Together, these objectives establish an integrated analytical framework to examine how platform features and psychological responses jointly shape attitudinal loyalty toward Đậu Bạc on e-commerce platforms.

RESEARCH QUESTIONS

01

What are the demographic characteristics (age, gender, occupation) and shopping frequency of consumers who are interested in Đậu Bạc products on contemporary e-commerce platforms?



Which platform-related and psychological factors can be identified through factor analysis to represent the online shopping experience for handcrafted jewelry?

02

03

How can consumers of Đậu Bạc products be segmented through cluster analysis based on differences in perceived platform features and psychological responses?



To what extent do platform features (interactivity, informativeness, visual engagement, and ease of navigation) influence consumers' attitudinal loyalty toward Đậu Bạc products?

04

05

How do psychological responses (trust, convenience, enjoyment, and self-control) contribute to shaping purchase intentions and long-term loyalty toward local cultural goods?



RESEARCH STRUCTURE

- 01** Introduces the research context, conceptual foundation, research gaps, and research objectives.
- 02** Describes the research methodology, including data sources, measurement development, and analytical techniques.
- 03** Presents the empirical results and discusses managerial implications for traditional craft-makers and e-commerce platforms.

EXPECTED CONTRIBUTIONS

This study is expected to contribute to the literature and practice by highlighting how digital platforms can play an active role in revitalizing and sustaining traditional cultural values in contemporary consumption contexts.

From a theoretical perspective, the study extends the Online Customer Shopping Experience (OCSE) framework by situating it within the context of traditional artisanal and cultural heritage products. Rather than treating e-commerce merely as a transactional channel, the research conceptualizes digital platforms as cultural intermediaries that mediate how heritage-based products, such as *Đậu Bạc* silver jewelry, are perceived, experienced, and emotionally valued by consumers. In doing so, the study contributes to the growing body of research that views digital consumption not only as an economic activity, but also as a mechanism for preserving and reactivating intangible cultural heritage in modern society.

Methodologically, the study contributes by employing an integrated, multi-stage analytical approach combining descriptive analysis, exploratory factor analysis, cluster analysis, and multiple regression. This framework enables a systematic examination of how platform features and psychological responses jointly shape attitudinal loyalty toward cultural products, thereby offering a structured way to translate abstract cultural values into measurable experiential and behavioral outcomes.

From a practical perspective, the findings provide meaningful guidance for artisans, cultural brands, and e-commerce practitioners seeking to bring traditional craftsmanship into the digital era without diluting its cultural essence. By identifying the experiential and psychological factors that strengthen loyalty, the study offers insights into how digital design, trust-building mechanisms, and emotionally engaging storytelling can help traditional products not only survive, but thrive, in online marketplaces. Ultimately, the research underscores the potential of digital platforms as strategic tools for sustaining, promoting, and renewing traditional cultural heritage in a rapidly evolving digital economy.

RESEARCH DESIGN & METHODOLOGY

Measurement and Variables

All variables in this study were measured using a 5-point Likert scale, ranging from 1 (“Strongly Disagree”) to 5 (“Strongly Agree”):

Table 1: Measurement and Variables

Construct	Code	Measurement Description	Role
Interactivity	INT1 – INT2	Measures the extent to which users can communicate, exchange opinions with other users, and the importance of peer advice during online shopping.	IV
Informativeness	INF1 – INF3	Assesses the completeness, level of detail, and perceived truthfulness of product information provided on the platform.	IV
Visual Engagement	VE1 – VE3	Measures user satisfaction with smart and intuitive design, color schemes, image quality, and typography of the platform.	IV
Navigation Ease	NVSE1 – NVSE2	Evaluates the ease of searching for products, navigating similar items, and accessing personalized product recommendations.	IV
Trust	TRUST1 – TRUST3	Measures users’ confidence in the platform’s reputation, perceived security, and sense of safety during online transactions.	IV
Convenience	CONV1 – CONV4	Assesses the ability to shop anytime and anywhere, and the extent to which online shopping saves time and effort.	IV

RESEARCH DESIGN & METHODOLOGY

Enjoyment	ENJ1 – ENJ3	Measures the degree of pleasure, excitement, and comfort experienced during the online shopping process, including engagement with promotional programs.	IV
Self-Control	SC1 – SC2	Evaluates consumers’ tendency to consider long-term value, avoid impulsive purchases, and reconsider orders before completion.	IV
Attitudinal Loyalty	AL1 – AL3	Measures consumers’ intention to prioritize the platform, perceived wisdom of online purchase decisions, and willingness to recommend the platform to others.	DV

Data Source and Data Collection

The primary data for this study were obtained from a structured survey-based quantitative dataset provided by the course instructor, ensuring academic consistency and methodological alignment with the course requirements. The dataset comprises responses from consumers who have practical experience in discovering and purchasing creative traditional and local artisanal products—specifically within the context of *Đậu Bọc*—via major Vietnamese e-commerce platforms.

The dataset is cross-sectional in nature, capturing consumer perceptions and evaluations at a single point in time. This design is appropriate for examining the relationships between platform features, psychological responses, and attitudinal loyalty within the online shopping environment.

Measurement Framework

Data were collected through a structured questionnaire designed to operationalize key constructs of the online customer shopping experience (OCSE). Measurement items were adapted from established scales in prior e-commerce and consumer behavior studies and refined to suit the context of traditional artisanal products. The framework comprises four platform features (Interactivity, Informativeness, Visual Engagement, and Navigation Ease) and four psychological responses (Trust, Convenience, Enjoyment, and Self-Control), all measured using a 5-point Likert scale (1 = “Strongly Disagree”, 5 = “Strongly Agree”).

After screening for completeness and consistency, 351 valid responses were retained for analysis. The finalized dataset was prepared for descriptive analysis, measurement validation, factor analysis, customer segmentation, and regression modeling.

RESEARCH DESIGN & METHODOLOGY

Overview of Survey Sections

The questionnaire used in this study was systematically structured to ensure logical flow and alignment with the research objectives. It consisted of four main sections, each designed to capture a specific group of variables relevant to consumers' online shopping behavior on Vietnamese e-commerce platforms.

The first section collected demographic information, including age, gender, occupation, and shopping frequency. These variables were used to describe the sample characteristics and to support group comparison analyses such as t-tests and ANOVA. They were not included in factor analysis, clustering, or regression models.

The second section focused on platform features, which represent consumers' functional and experiential evaluations of e-commerce platforms. This section measured interactivity, informativeness, visual engagement, and ease of navigation. These constructs capture how platform design, information quality, and usability shape consumers' online shopping experiences.

The third section measured psychological factors, including trust, convenience, enjoyment, and self-control. These variables reflect consumers' internal cognitive and emotional responses during online shopping and serve as key explanatory mechanisms linking platform features to attitudinal outcomes.

The final section assessed attitudinal loyalty, capturing consumers' psychological commitment and favorable attitudes toward purchasing traditional and creative local products on e-commerce platforms. Attitudinal loyalty was treated as the primary outcome variable in subsequent regression and profiling analyses.

Relevance of Questionnaire Design

The structure and content of the questionnaire were designed to reflect the characteristics of the fashion-oriented and artisanal product context, where visual appeal, emotional engagement, and trust play a particularly important role in shaping consumer behavior. Traditional and creative local fashion products often rely on visual presentation, storytelling, and perceived authenticity, making constructs such as visual engagement and trust especially relevant.

In addition, psychological factors such as enjoyment and self-control are closely associated with fashion-related online shopping, where hedonic motivation and impulse regulation frequently influence purchase decisions. By organizing the questionnaire around platform features, psychological responses, and attitudinal loyalty, the survey design provides a coherent framework for examining how digital platform experiences translate into loyalty toward traditional and creative local fashion products in an e-commerce environment.

RESEARCH DESIGN & METHODOLOGY

Analytical Tools and Software

This study employed Python as the primary analytical tool within the Google Colab environment to conduct quantitative data analysis. Python was selected for its flexibility, reproducibility, and suitability for handling large-scale survey data.

Data preprocessing and transformation were performed using the Pandas and NumPy libraries. Descriptive statistics, correlation analysis, group comparisons (T-test and ANOVA), and multiple regression models were conducted using SciPy and statsmodels. Measurement quality and construct validity were assessed through Exploratory Factor Analysis (EFA), including Kaiser–Meyer–Olkin (KMO) and Bartlett’s Test of Sphericity, implemented via the factor_analyzer package.

Customer segmentation was carried out using K-means clustering with support from scikit-learn, while data visualization (bar charts, correlation heatmaps, scatter plots, and radar charts) was generated using Matplotlib and Seaborn to enhance interpretability.

All analyses were conducted in a reproducible computational environment to ensure analytical rigor, consistency, and transparency.

Analytical Procedures

Phase 1: Data Screening and Descriptive Profiling



The objective of this phase is to ensure data quality and to provide an overview of the sample characteristics prior to hypothesis testing. Data screening procedures, including duplicate removal and missing value handling, were applied, followed by descriptive statistical analyses using frequency, percentage, and mean values.

(1) Mean (Average value):

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The mean represents the central tendency of respondents' evaluations for each variable.

(2) Percentage:

$$\text{Percentage} = \frac{f_i}{N} \times 100\%$$

Percentage indicates the relative proportion of each category within the sample.

This phase is descriptive in nature and does not involve hypothesis testing.

RESEARCH DESIGN & METHODOLOGY

Analytical Procedures

Phase 2: Measurement Quality Assessment



This phase aims to evaluate the adequacy, validity, and internal structure of the measurement scales prior to factor extraction. The analysis employs the Kaiser–Meyer–Olkin (KMO) measure, Bartlett’s Test of Sphericity, and Exploratory Factor Analysis (EFA)..

(1) Kaiser–Meyer–Olkin (KMO) Measure:

$$KMO = \frac{\sum \sum r_{ij}^2}{\sum \sum r_{ij}^2 + \sum \sum q_{ij}^2}$$

The KMO index assesses sampling adequacy and indicates whether factor analysis is appropriate.

KMO > 0.60 indicates sufficient suitability for EFA.

(2) Bartlett’s Test of Sphericity:

$$\chi^2 = - \left(n - 1 - \frac{2p + 5}{6} \right) \ln |R|$$

This test examines whether the correlation matrix significantly differs from an identity matrix.

p-value < 0.05 indicates that variables are sufficiently correlated.

(3) Eigenvalue (Kaiser Criterion):

$$\lambda_k > 1 \Rightarrow \text{Retain factor}$$

Eigenvalues represent the amount of variance explained by each factor.

(4) Factor Loading Model:

$$\text{Cumulative Variance} = \frac{\sum_{k=1}^m \lambda_k}{\sum \lambda}$$

Explained variance reflects the proportion of total variance captured by retained factors.

Phase 3: Customer Segmentation (Cluster Analysis)



This phase aims to identify homogeneous customer groups based on similarities in perceptions and behavioral tendencies. Z-score standardization was applied prior to K-means clustering, with the optimal number of clusters determined using the Elbow method.

(1) Z-score Standardization:

$$z = \frac{x - \mu}{\sigma}$$

Standardization ensures all variables contribute equally to the clustering process.

(2) K-means Objective Function:

$$\min \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

The algorithm minimizes within-cluster variance.

RESEARCH DESIGN & METHODOLOGY

Analytical Procedures

Phase 3: Customer Segmentation (Cluster Analysis)

(3) Within-Cluster Sum of Squares (WCSS):

$$WCSS = \sum \|x_i - \mu_k\|^2$$

WCSS measures cluster compactness.

The Elbow point indicates the optimal number of clusters.

Phase 4: Group Comparison (T-Test & ANOVA)

This phase aims to examine statistically significant differences across demographic groups using Independent Samples T-Test and One-way ANOVA, with Levene's Test applied to assess the assumption of variance homogeneity.

(1) Independent Samples T-Test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

This test compares mean differences between two independent groups.

(2) Levene's Test

$$W = \frac{(N - k)}{(k - 1)} \cdot \frac{\sum n_j (\bar{Z}_j - \bar{Z})^2}{\sum \sum (Z_{ij} - \bar{Z}_j)^2}$$

Levene's test evaluates the homogeneity of variances assumption.

(3) One-way ANOVA

$$F = \frac{MS_{between}}{MS_{within}}, \quad \text{where } MS = \frac{SS}{df}$$

ANOVA assesses mean differences across more than two groups.

RESEARCH DESIGN & METHODOLOGY

Analytical Procedures

Phase 5: Correlation and Multicollinearity Diagnostics



This phase aims to ensure that independent variables do not distort regression estimates by examining multicollinearity using Pearson correlation coefficients, Variance Inflation Factor (VIF), and the Condition Index.

(1) Pearson Correlation

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Pearson's r measures the strength and direction of linear relationships.

(2) Variance Inflation Factor (VIF)

$$\text{VIF}_i = \frac{1}{1 - R_i^2}$$

VIF quantifies multicollinearity among predictors.

$\text{VIF} < 5$ indicates acceptable levels.

(3) Condition Index

$$\text{CI} = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}}$$

The condition index detects potential multicollinearity issues.

Phase 6: Determinants Analysis (Regression Models)



This phase aims to identify the most influential determinants of Attitudinal Loyalty using Multiple Linear Regression (OLS).

(1) Regression Model

$$AL = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \varepsilon$$

This model estimates the direct impact of independent variables on attitudinal loyalty.

(2) Ordinary Least Squares (OLS) Estimation

$$\hat{\beta} = (X'X)^{-1}X'Y$$

OLS estimates regression coefficients by minimizing the sum of squared residuals.

RESEARCH RESULTS & ANALYSES

Descriptive Analysis (RQ1)

Purpose of this section: This section presents the descriptive statistics of the demographic characteristics of the respondents to address Research Question 1 (RQ1).

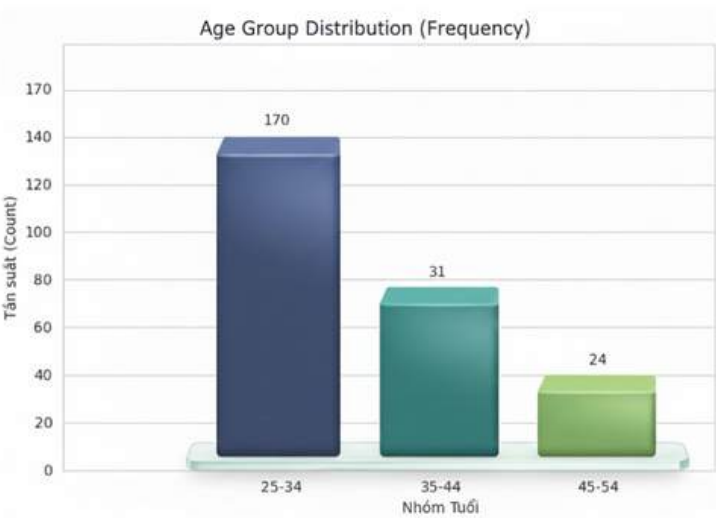


Figure 1: Age group distribution chart

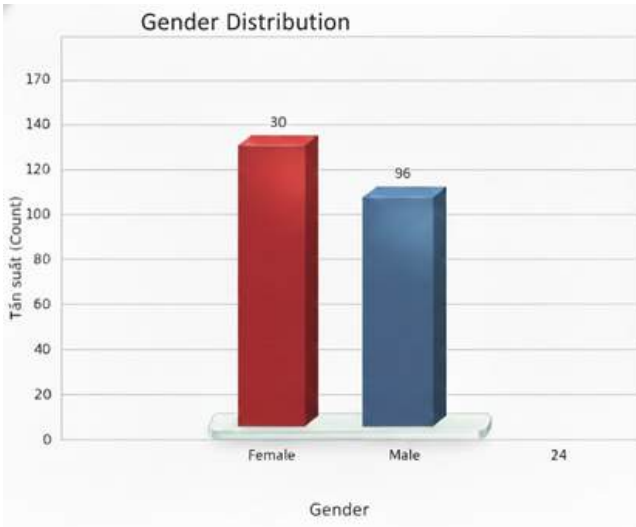


Figure 2: gender distribution char

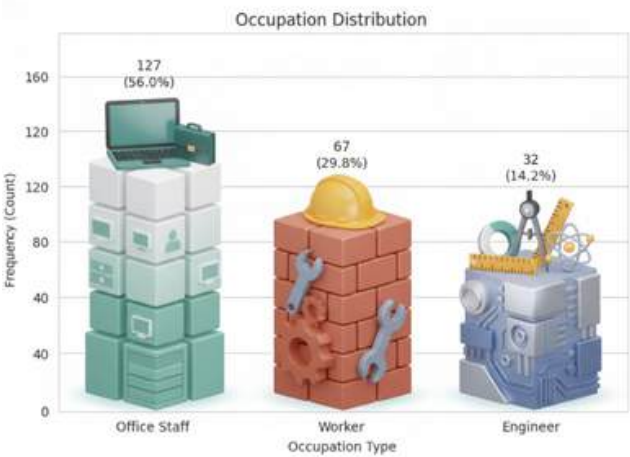


Figure 3: Occupation distribution chart

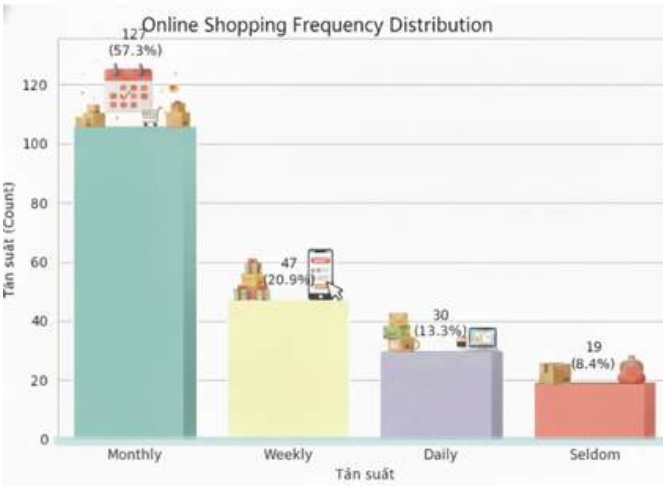


Figure 4: frequency distribution chart

RESEARCH RESULTS & ANALYSES

Descriptive Analysis (RQ1)

Age group distribution :

The 25-34 age group is the largest, with 170 respondents (57.3%), indicating that traditional products attract this demographic, especially those in their professional or established career stage. The 35-44 (13.3%) and 45-54 (8.4%) age groups are smaller but still significant, potentially interested in the product for its cultural value and uniqueness. Marketing should focus on the 25-34 age group, while also targeting the older groups with messages about quality and tradition.

Gender distribution:

The Male group is dominant, with 96 respondents (76.2%) compared to Female (23.8%), indicating that this traditional product has a stronger appeal to men. This likely reflects a greater interest from male consumers in culturally significant products. To reach female consumers, marketing strategies could emphasize aesthetics and household applications.

Occupation distribution:

The Office Staff group is the largest, with 127 respondents (56.0%), showing that traditional products are particularly appealing to white-collar workers. The Worker and Engineer groups, though smaller, still express interest in the product, especially for its quality and sustainability. Marketing strategies should emphasize practicality for the Worker group and craftsmanship for the Engineer group, while also highlighting cultural value for Office Staff.

Occupation distribution:

The Monthly group is the largest, with 127 respondents (57.3%), indicating that most consumers shop online on a monthly basis. The Weekly (20.9%) and Daily (13.3%) groups show strong interest in frequently shopping for new products or promotions. Marketing strategies should focus on quality and value for the Monthly group, while frequently updating products and offering promotions for the more frequent shoppers.



RESEARCH RESULTS & ANALYSES

Measurement Quality Assessment

Table 2: KMO and Bartlett's Test Results

Measure	Value
Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy	0.8557
Bartlett's Test of Sphericity (Approx. Chi-Square)	5509.72
Bartlett's Test of Sphericity (Sig.)	0

The KMO value of 0.8557 (considered "Great") and a Bartlett's Test significance of 0.000 confirm that the sample size is highly adequate and the data is statistically suitable for factor analysis.

Table 3: Total Variance Explained (Initial Eigenvalues)

Factor	Initial Eigenvalues	% of Variance	Cumulative %
Factor 1	3.726	14.90%	14.90%
Factor 2	3.6587	14.63%	29.54%
Factor 3	2.0242	8.10%	37.64%
Factor 4	2.0139	8.06%	45.69%
Factor 5	1.5387	6.15%	51.85%
Factor 6	1.4694	5.88%	57.72%
Factor 7	1.2186	4.87%	62.60%
Factor 8	0.9984	3.99%	66.59%
Factor 9	0.6616	2.65%	69.24%

REASONS FOR CHOOSING 9 FACTORS :

1. Data Suitability:

The data is highly suitable for factor analysis with a KMO index of 0.8557 (considered "Great"). Bartlett's Test of Sphericity is statistically significant ($p = 0.000$, Chi-square = 5509.72), confirming strong correlations between the observed variables.

RESEARCH RESULTS & ANALYSES

Measurement Quality Assessment

2. Explanatory Power and Extraction :

The model extracts 9 factors, accounting for 69.24% of the total variance explained. Selecting 9 factors optimizes information retention (nearly 70%) and ensures comprehensive coverage of Đậu Bạc's specific marketing dimensions.

3. Scale Convergence :

The Rotated Factor Matrix shows strong convergence, with most Factor Loadings exceeding 0.50. Representative variables such as sc1 (0.941), ve3 (0.890), and nvse2 (0.883) confirm a robust link between the survey data and the underlying factor structure.

4. Reliability :

Overall, the 9-factor model demonstrates high convergent and discriminant validity. This provides a solid and reliable database for Đậu Bạc to proceed with subsequent quantitative analysis and customer segmentation.

Nine extracted factors account for 69.24% of the total variance, exceeding the 60% threshold required in social science research. This indicates that the measurement model provides a comprehensive representation of the original variables regarding the Đậu Bạc e-commerce experience.

Table 4: KMO and Bartlett's Test Results

Item	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7
int1	0.562						
int2	0.75						
inf1				0.533			
inf2				0.77			
inf3				0.734			
ve1			-0.41				
ve2			-0.466		0.542		
ve3					0.89		
nvse1	-0.681						
nvse2	-0.425						
trust1	0.425						0.581
trust2		0.526					0.626
trust3			0.551				
conv1	0.422	0.439					
conv2	0.486	0.554					
conv3	0.426	0.571					
conv4		0.609					
enj1		0.706					
enj2	0.538						
enj3	0.675						
sc1						0.941	
sc2						0.643	
al1		-0.541	-0.593				
al2		-0.691					
al3		-0.429	-0.597				

Overall, the rotated factor matrix demonstrates clear factor separation, strong factor loadings (≥ 0.40), and satisfactory convergent validity across constructs. The extracted factors provide a statistically sound measurement structure, forming a reliable foundation for subsequent cluster analysis and regression modeling.

RESEARCH RESULTS & ANALYSES

Customer Segmentation (Cluster Analysis)

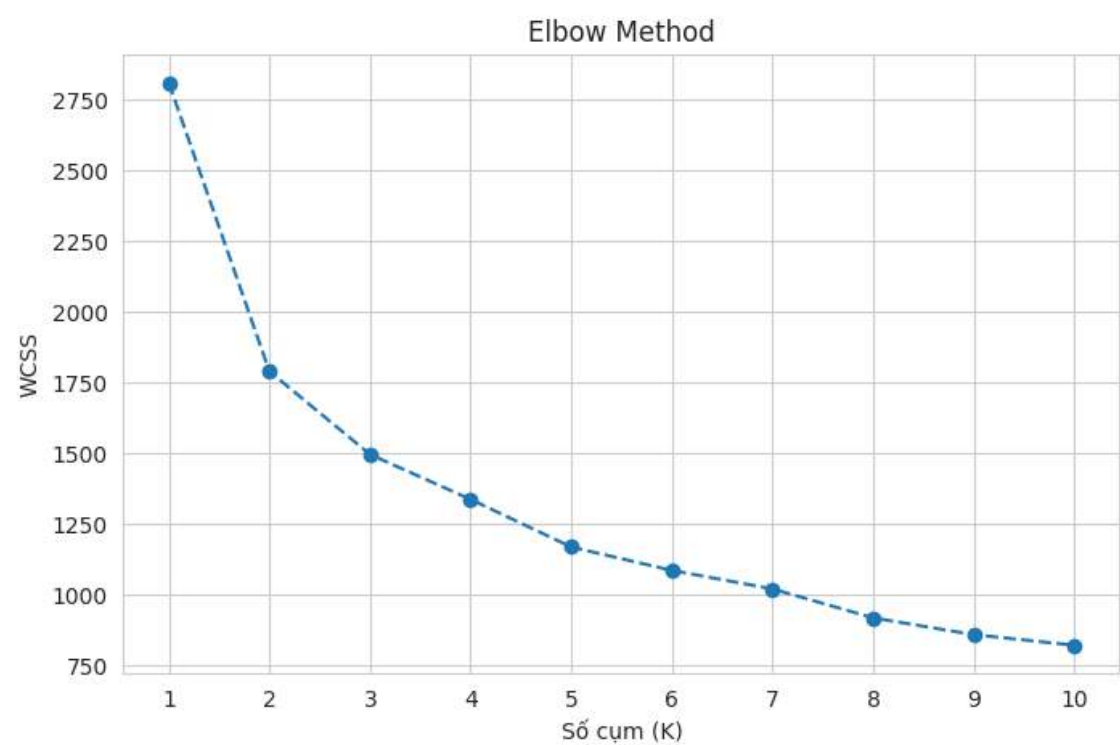


Figure 5: Elbow Method for Determining the Optimal Number of Clusters

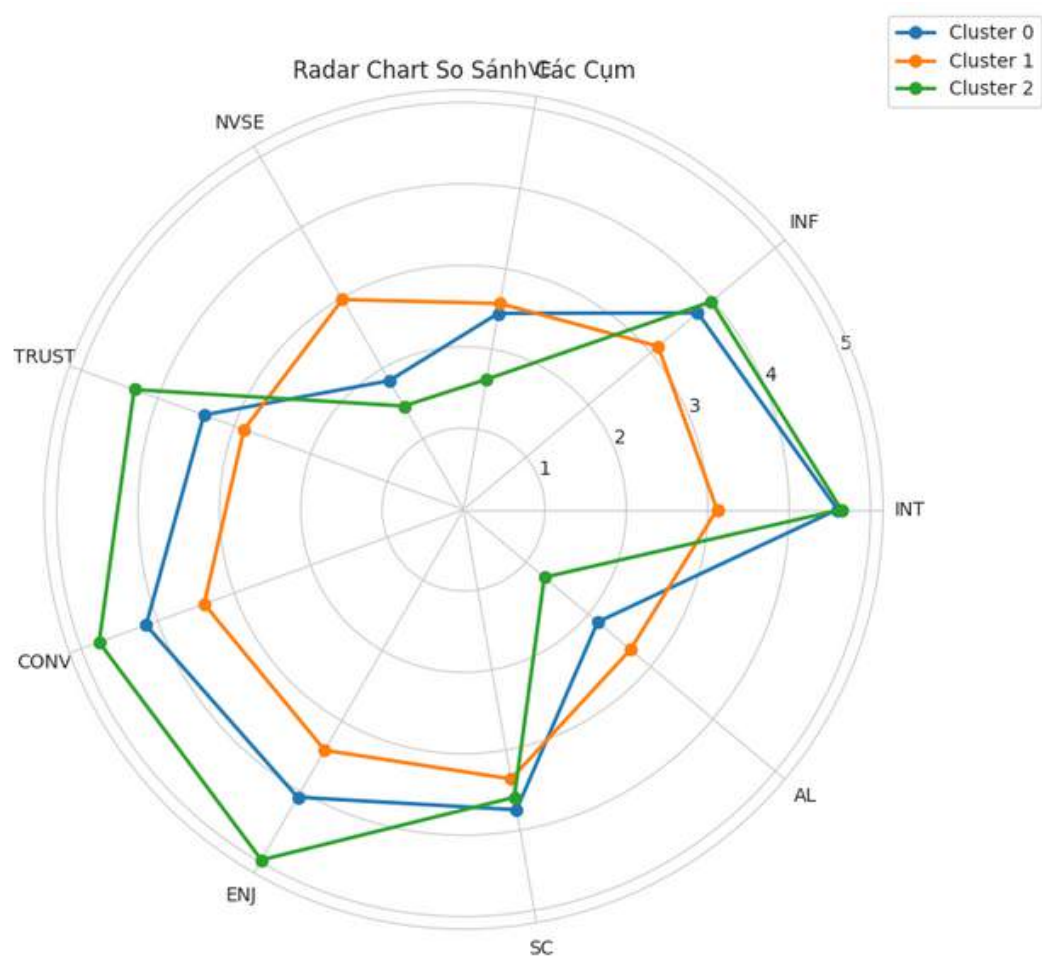


Figure 6: Radar Chart Comparing Cluster Profiles

RESEARCH RESULTS & ANALYSES

Customer Segmentation (Cluster Analysis)

To examine heterogeneity in consumers' online shopping experiences, a K-means cluster analysis was conducted using the mean scores of nine validated constructs: Interactivity (INT), Informativeness (INF), Visual Engagement (VE), Navigation Ease (NVSE), Trust (TRUST), Convenience (CONV), Enjoyment (ENJ), Self-Control (SC), and Attitudinal Loyalty (AL). All construct scores were standardized prior to analysis to ensure equal contribution across variables. Based on the elbow criterion, a three-cluster solution was selected as it offers an appropriate balance between parsimony and explanatory power.

The results reveal three distinct consumer segments with clearly differentiated patterns of platform evaluation and attitudinal loyalty toward traditional artisanal products on e-commerce platforms. The relative sizes of the clusters indicate meaningful behavioral diversity, with no single segment dominating the sample.

Cluster 1: Functional and Moderately Loyal Pragmatists (39.89%)

Consumers in this segment exhibit moderate and relatively balanced evaluations across platform features and psychological responses. Navigation Ease and Convenience are perceived more favorably than experiential attributes, while Trust and Enjoyment remain at moderate levels. Despite not exhibiting the strongest experiential engagement, this cluster reports the highest Attitudinal Loyalty. This pattern suggests that loyalty in this segment is driven primarily by functional reliability, usability, and habitual usage rather than emotional immersion.

Cluster 2: Experience-Oriented but Low-Loyalty Users (37.89%)

This cluster reports high evaluations of experiential and functional dimensions, including Interactivity, Informativeness, Convenience, Enjoyment, and above-average Trust. However, Attitudinal Loyalty remains relatively low compared to the pragmatist segment. The findings indicate that although these consumers actively enjoy the shopping experience, such enjoyment does not necessarily translate into long-term attitudinal commitment. They are likely to switch platforms when comparable alternatives are available.

Cluster 3: High-Experience but Autonomy-Driven Consumers (22.22%)

Consumers in this segment demonstrate the highest levels of Trust, Convenience, Enjoyment, and Self-Control, reflecting strong confidence in platform performance and high shopping satisfaction. Nevertheless, this group exhibits the lowest Attitudinal Loyalty. This paradox highlights an autonomy-driven profile in which strong self-control and rational evaluation reduce emotional dependence on any single platform, even when experiences are highly positive.

Interpretation of Segmentation Results

Overall, the cluster analysis reveals substantial heterogeneity in how consumers experience and respond to e-commerce platforms. The findings show that superior platform performance and enjoyable experiences do not automatically result in higher attitudinal loyalty. Instead, loyalty formation depends on a complex interplay between functional reliability, experiential engagement, and consumers' psychological orientation toward autonomy and self-control. These results underscore the importance of adopting segment-specific strategies when marketing traditional artisanal products such as Đậu Bọc in digital environments.

RESEARCH RESULTS & ANALYSES

Group Comparison (T-Test & ANOVA)

Gender Differences (Independent Samples T-Test)

An independent samples t-test was conducted to assess gender differences in attitudinal loyalty. Levene's test indicated unequal variances (Levene's Statistic = 4.3176, $p = 0.0403$); therefore, Welch's t-test was applied. The results reveal a statistically significant difference in attitudinal loyalty between male and female respondents ($t = 2.9543$, $p = 0.0039$). Male consumers reported higher attitudinal loyalty ($M = 5.7646$) than female consumers ($M = 5.0552$), indicating that gender plays a meaningful role in shaping loyalty toward Đậu Bạc on e-commerce platforms.

Age Group Comparison (One-Way ANOVA)

A one-way ANOVA was conducted to examine differences across age groups. The results indicate a statistically significant difference for Interactivity (INT) ($F = 3.656$, $p = 0.0268$), suggesting that consumers from different age cohorts value interactive platform features differently. No significant differences were found for other constructs, including Attitudinal Loyalty ($p > 0.05$). Overall, age influences interaction perceptions but does not significantly affect loyalty toward Đậu Bạc.

Occupational Group Comparison (One-Way ANOVA)

The one-way ANOVA results show no statistically significant differences across occupational groups for any platform features, psychological responses, or attitudinal loyalty ($p > 0.05$). This suggests that occupational background does not substantially influence consumers' evaluations or loyalty toward traditional artisanal products.

Shopping Frequency Comparison (One-Way ANOVA)

A further one-way ANOVA indicates a statistically significant difference for Interactivity (INT) across shopping frequency groups ($F = 2.643$, $p = 0.0492$). More frequent shoppers tend to place greater emphasis on interactive functions such as reviews and peer communication. However, no significant differences were observed for Attitudinal Loyalty or other constructs ($p > 0.05$), implying that shopping frequency affects interaction expectations rather than long-term loyalty.

Summary of Group Comparison Findings

Overall, demographic factors exert selective effects on consumer behavior. Gender is the only variable with a direct and significant impact on attitudinal loyalty, while age and shopping frequency primarily influence perceptions of interactivity. Occupational differences are negligible. These findings reinforce the conceptual framework, suggesting that attitudinal loyalty toward traditional artisanal products is driven less by demographics and more by experiential and psychological mechanisms embedded in the e-commerce platform.

RESEARCH RESULTS & ANALYSES

Brief explanation of the plots

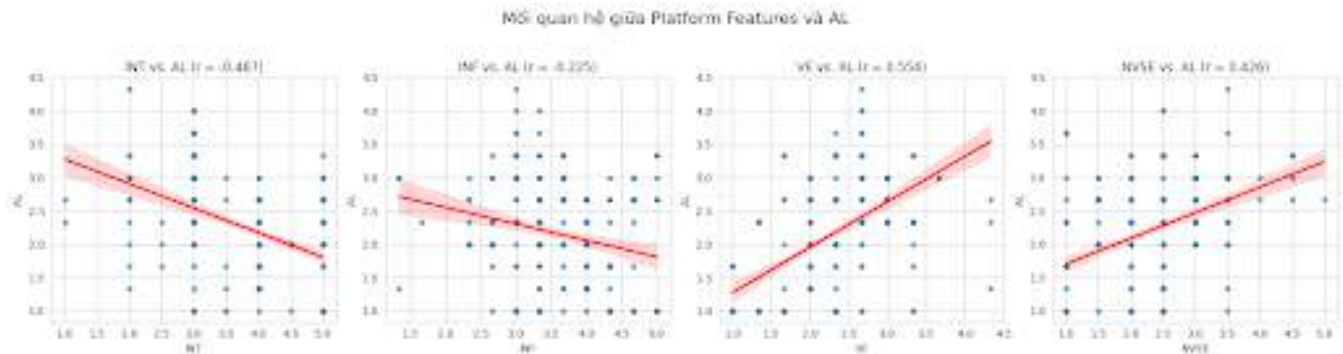


Figure 7: Scatter Plot: Platform Features vs. Attitudinal Loyalty (AL)

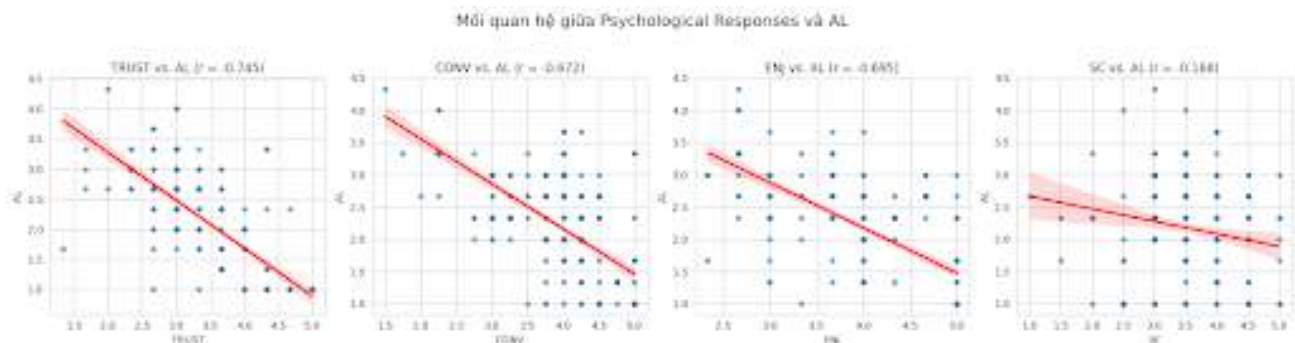


Figure 8: Scatter Plot: Psychological Responses vs. Attitudinal Loyalty (AL)

The scatter/regplots illustrate the bivariate linear relationship between each determinant and Attitudinal Loyalty (AL):

- Each dot represents one respondent.
- The red line is the fitted linear trend (linear fit).
- The light red shaded area shows the confidence interval around the fitted line.
- Because the data are based on a Likert scale (1–5), points often appear in vertical “bands” at the same score values, which is normal.

For each subplot, the Pearson correlation coefficient (r) is displayed in the title. A larger $|r|$ indicates a stronger linear relationship, while the sign (+/–) shows the direction of the relationship.

RESEARCH RESULTS & ANALYSES

Results and interpretation

Pearson correlation analysis was conducted to examine the bivariate linear relationships between platform features (INT, INF, VE, NVSE), psychological responses (TRUST, CONV, ENJ, SC), and Attitudinal Loyalty (AL). Correlation reflects association rather than causation; therefore, the results indicate patterns observed in the sample, not evidence that one variable directly causes changes in another.

A) Platform Features vs. Attitudinal Loyalty (Figure 7)

Overall, AL shows statistically significant relationships with all four platform-feature variables, but the strength varies considerably.

- Visual Engagement (VE) → AL: $r = 0.554$ (positive, moderate-to-strong)

VE demonstrates the strongest positive association with AL among platform features. This suggests that respondents who rate the platform's visual presentation higher (e.g., design, image quality, interface aesthetics) tend to report higher attitudinal loyalty. In the context of traditional artisanal products—where visual cues often communicate craftsmanship, authenticity, and cultural value—this pattern is theoretically plausible.

- Ease of Navigation (NVSE) → AL: $r = 0.426$ (positive, moderate)

NVSE also shows a meaningful positive relationship with AL, indicating that smoother navigation and easier product discovery are associated with higher loyalty attitudes. This implies that reducing friction in searching, filtering, and finding relevant artisanal items may support stronger loyalty perceptions.

- Interactivity (INT) → AL: $r = -0.467$ (negative, moderate)

INT exhibits a moderate negative association with AL. This pattern suggests that higher perceived interactivity is associated with lower loyalty attitudes in this dataset. One plausible interpretation is that certain “interactive” elements (e.g., excessive prompts, interruptions, or high-effort interactions) may be perceived as intrusive or cognitively demanding, thereby weakening users' loyalty attitudes rather than strengthening them.

- Informativeness (INF) → AL: $r = -0.225$ (negative, weak)

INF shows a weak negative association with AL. This indicates that greater perceived informativeness does not necessarily translate into stronger loyalty attitudes and may even slightly coincide with lower AL. This could reflect information overload, credibility concerns, or a mismatch between “more information” and “more meaningful value” in user evaluation.

Within platform features, VE and NVSE emerge as the most supportive correlates of AL, while INT and INF show negative associations, suggesting that “more interactive/more informational” is not automatically better for building loyalty in this sample.

RESEARCH RESULTS & ANALYSES

Results and interpretation

B) Psychological Responses vs. Attitudinal Loyalty (Figure 8)

Psychological response variables demonstrate substantially stronger absolute correlations with AL than platform features, highlighting that loyalty attitudes are more tightly linked to internal perceptions (trust, enjoyment, convenience) than to surface platform characteristics.

- TRUST → AL: $r = -0.745$ (very strong in magnitude)
- ENJ → AL: $r = -0.695$ (strong in magnitude)
- CONV → AL: $r = -0.672$ (strong in magnitude)
- These three variables show the largest absolute correlations with AL, indicating that attitudinal loyalty is most strongly associated with core psychological evaluations.
- Self-Control (SC) → AL: $r = -0.168$ (weak in magnitude)
- SC is only weakly associated with AL, suggesting it plays a smaller role relative to trust, enjoyment, and convenience in shaping loyalty attitudes.

The negative signs for TRUST, ENJ, and CONV may appear counterintuitive because marketing theory typically expects these constructs to align positively with loyalty. A likely methodological explanation is coding direction or reverse-coded items (e.g., higher numeric values representing lower trust/enjoyment/convenience, or AL being reverse-scored). Therefore, interpretation should prioritize association strength ($|r|$) and verify the coding scheme before drawing strong directional managerial conclusions.

Regardless of sign, the magnitude indicates that TRUST, ENJ, and CONV are the dominant correlates of AL and should be treated as key mechanisms in subsequent multivariate modeling.

=> Overall, attitudinal loyalty is much more strongly associated with psychological responses (TRUST, ENJ, CONV) than with platform features.

Within platform features, visual engagement and ease of navigation show the most consistent positive links with loyalty.



RESEARCH RESULTS & ANALYSES

Key takeaways from the plots

The plots and correlation coefficients indicate that the psychological response variables (especially TRUST, ENJ, and CONV) are much more strongly associated with AL than most platform-related factors. Within the platform features, Visual Engagement (VE) stands out as the strongest positive factor, followed by NVSE.

Note on the sign (+/-): The sign of r depends on how the Likert-scale items were coded (including any reverse-coded items). Therefore, this section emphasizes relationship strength ($|r|$), the direction of the fitted trend lines, and high–low group differences; the directional interpretation should be checked against the final coding scheme before drawing managerial conclusions.

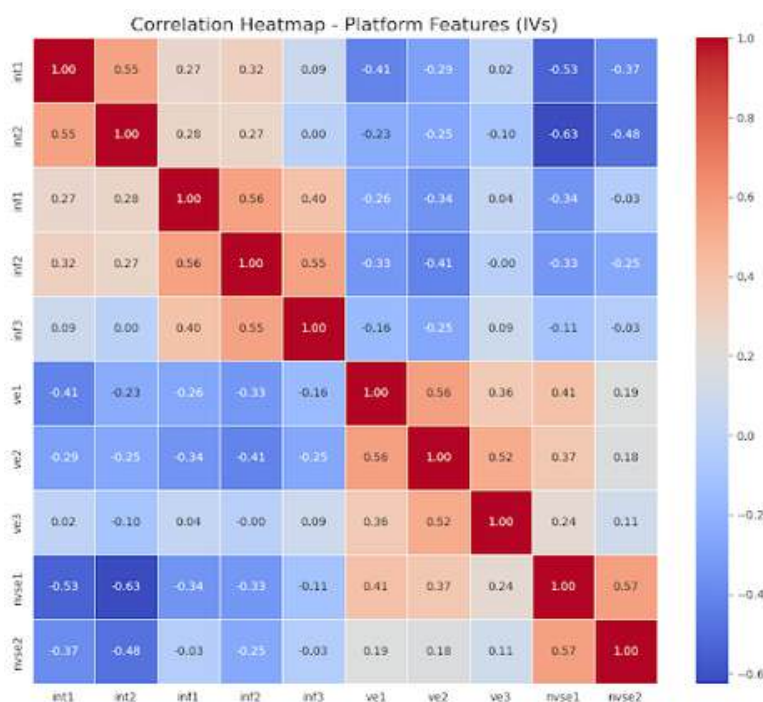


Figure 9: Correlation heatmap - Platform Features

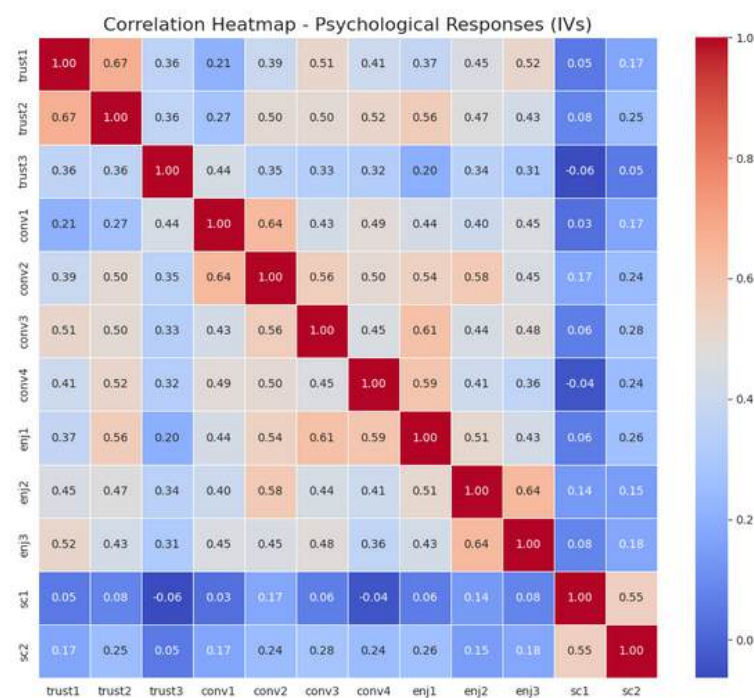


Figure 10: Correlation heatmap - Psychological Responses

Determinants Analysis

This section applies multiple regression to identify which factors are most strongly associated with Attitudinal Loyalty (AL) when all determinants are considered simultaneously. Unlike correlation analysis (Section 3.5), regression estimates the unique contribution of each predictor to AL while holding the others constant. The dependent variable is AL, computed as the mean of $al1$ – $al3$. The independent variables include four platform features—INT = mean(int1, int2), INF = mean(inf1, inf2, inf3), VE = mean(ve1, ve2, ve3), NVSE = mean(nvse1, nvse2)—and four psychological responses—TRUST = mean(trust1, trust2, trust3), CONV = mean(conv1, conv2, conv3, conv4), ENJ = mean(enj1, enj2, enj3), SC = mean(sc1, sc2).

RESEARCH RESULTS & ANALYSES

Determinants Analysis

The general regression model is:

$$AL_i = \beta_0 + \beta_1 INT_i + \beta_2 INF_i + \beta_3 VE_i + \beta_4 NVSE_i + \beta_5 TRUST_i + \beta_6 CONV_i + \beta_7 ENJ_i + \beta_8 SC_i + \varepsilon_i$$

Because multiple predictors were entered simultaneously, multicollinearity was assessed to ensure stable and interpretable coefficient estimates. Variance Inflation Factor (VIF) values were first examined using conventional thresholds (e.g., $VIF < 5$), indicating no severe multicollinearity among predictors. The Condition Index (CI) was then evaluated as a complementary diagnostic; all CI values fell within acceptable ranges, suggesting the absence of problematic collinearity patterns. Therefore, regression coefficients can be interpreted as reflecting each predictor's unique contribution to Attitudinal Loyalty (AL).

The regression model demonstrates strong explanatory power, with $R^2 \approx 0.655$, indicating that approximately 65% of the variance in AL is explained by the included platform and psychological determinants. This suggests that the selected predictors jointly account for a substantial share of variation in attitudinal loyalty.

The results reveal clear differences in predictor importance once all variables are considered simultaneously. Visual Engagement (VE) emerges as the strongest statistically significant determinant of AL ($\beta \approx 0.455$; $p < 0.001$), indicating that higher perceived visual appeal and presentation quality are strongly associated with greater attitudinal loyalty. Trust (TRUST) is also a significant predictor ($\beta \approx -0.383$; $p < 0.001$), confirming the central role of credibility-related perceptions in shaping loyalty. In contrast, Convenience (CONV) is not statistically significant in the full model ($p > 0.05$), implying limited unique explanatory power once other psychological factors are controlled for.

Standardized beta coefficients further confirm that VE is the most influential determinant of AL, followed by TRUST, while CONV does not function as an independent driver in the multivariate specification. Overall, although several variables may correlate with AL at the bivariate level, VE and TRUST remain the most robust and influential determinants in the regression model.

From a managerial perspective, the findings suggest prioritizing improvements in visual engagement—such as high-quality imagery, clear layouts, and aesthetic design—and strengthening trust through transparent information, authentic reviews, reliable delivery, and responsive customer support. Convenience remains relevant as part of the overall experience but does not independently drive attitudinal loyalty when other psychological perceptions are considered simultaneously.

These results directly support the study's objectives by identifying key determinants of attitudinal loyalty and clarifying which experience dimensions are most strongly aligned with loyalty outcomes in this e-commerce context.

INSIGHT FOR ĐẬU BẠC

HOW DIGITAL EXPERIENCE DRIVES ATTITUDINAL LOYALTY FOR TRADITIONAL SILVER JEWELRY

Đậu Bạc should compete through visual storytelling and cultural authenticity, transforming its e-commerce presence into a digital extension of traditional craftsmanship rather than a promotion-driven storefront.



Rather than competing on price or frequency-based promotions, Đậu Bạc should compete on meaning—where visual experience and cultural authenticity become the core sources of differentiation on e-commerce platforms.

VISUAL ENGAGEMENT AND TRUST—NOT DEMOGRAPHICS—ARE THE STRONGEST DRIVERS OF LOYALTY TOWARD ĐẬU BẠC ON E-COMMERCE PLATFORMS.



LATEST ARTICLE



WHO IS THE CORE CUSTOMER?

Young and middle-aged adults (25–44), office workers, and occasional online shoppers. Position Đậu Bạc as a lifestyle-oriented artisanal brand, emphasizing cultural value and aesthetics rather than price-driven promotions.



WHAT DRIVES LOYALTY MOST?

Visual Engagement is the strongest determinant of Attitudinal Loyalty. Invest in professional imagery, close-up craftsmanship details, and culturally meaningful visual storytelling to strengthen emotional attachment.



WHY DO CUSTOMERS STAY LOYAL?

Trust and enjoyment matter more than functional convenience. Build trust through transparent information, authenticity cues, reviews, and clear return policies, while enhancing enjoyment via storytelling and limited collections.

STRATEGIC TAKEAWAY:

For traditional artisanal brands like Đậu Bạc, loyalty in digital environments is not earned through convenience or demographic targeting, but through the ability to transform heritage into visually compelling and trustworthy experiences that consumers choose to stay with.



SHOPEE STRATEGY FOR DAU BAC

Đậu Bạc leverages Shopee as a visual-first storytelling platform, using high-quality product images, close-up craftsmanship shots, and consistent aesthetics to build emotional attachment. Trust is reinforced through verified reviews, authenticity cues, and transparent product information, while lifestyle-oriented collections and limited designs enhance enjoyment and attitudinal loyalty beyond price competition.

Online 40 phút trước

Dau Bac Dinh Cong

Online 40 phút trước

Đang Theo: 8

Đánh Giá: 4.9 (12,7k Đánh Giá)

Ti Lệ Phản Hồi Chat: 100% (Trong Vài Giờ)

Tham Gia: 22 Tháng Trước

phụ kiện

bông tai

nhẫn

dây chuyền

Thêm

Giảm 10kđ

Đơn Tối Thiểu 180kđ

Sản phẩm nhất định

Lưu

HSD: 28.03.2025

GỢI Ý CHO BẠN

Xem Tất Cả >

Vòng tay bạc nữ Tyche Butterfly sang trọn...

179.100đ -10%

Lắc tay bạc Tyche Camellia charm nơ đỉnh ...

179.100đ -10%

Lắc Tay Bạc Nữ Tyche Black Leaf, Vòng Tay Cổ ...

180.000đ -10%

Lắc tay bạc nữ Tyche Blossom dây xù bắt sáng...

190.000đ -5%

Dây Chuyền Bạc Nữ Tyche Ocean Nam Đả...

179.100đ -10%

Lắc tay bạc nữ MERMAID AND THE SE...

175.500đ -10%

Shop

Sản phẩm

Danh mục hàng

5.0 Đánh giá shop

5.0 Đánh giá shop

5.0 Đán

Nhà Bạc Dinh Công - Fine Silver Fillrige Jewelry

ip Lotus Collection Now

BEST SELLERS

TOP 1

HOA SEN PEARLWOOD INDIGO GROUP7 MDA

450.000đ

TOP 2

HOA SEN PEARLWOOD INDIGO GROUP7 MDA

450.000đ

Shopee đã cung cấp các nhóm phân loại và giá trị phẩm loại đã được tiêu chuẩn hóa để giúp sản phẩm của bạn được tìm kiếm dễ dàng hơn.

Thần Ấp Việt

Tên sản phẩm *

Ghim cài áo Đậu Bạc Dinh Công

Mô tả sản phẩm *

Sinh ra từ kỹ nghệ thủ công bạc truyền thống của người Việt, Đậu Bạc không chạy theo sự phổ biến của kim loại quý, mà đi tìm vẻ đẹp bền bỉ của lao động, thời gian và kỹ óc. Mỗi hạt bạc nhỏ -đậu bạc- được rung chuông, se tay, gắn kết thủ công bằng sự kiên nhẫn gần như thiên định. Không có hai món trang sức nào hoàn toàn giống nhau, bởi mỗi sản phẩm mang theo nhịp thở của người thợ đã tạo nên nó. Trong một thế giới nơi mọi thứ có thể được sao chép chỉ bằng một cú click, Đậu Bạc chọn con đường ngược lại: chậm hơn, sâu hơn, và thật hơn.

Ngành hàng *

Tính chất đường >

GTIN >

Thương hiệu *

No brand >

Trọng lượng >

Bạc Dinh Công - Fine - Fine Silver Fillrige Jewelry

- Boat & Draggrooffly Collection Now

BEST SELLERS

TOP 1

HOA SEN PEARLWOOD INDIGO GROUP7 MDA

450.000đ

TOP 2

HOA SEN PEARLWOOD INDIGO GROUP7 MDA

450.000đ

CONCLUSION

This study examined how digital experience shapes attitudinal loyalty toward traditional artisanal products, using Đậu Bạc as an empirical case on Vietnamese e-commerce platforms. Consistent with prior research on online customer shopping experience, the findings demonstrate that customer loyalty is driven primarily by experiential and psychological mechanisms rather than demographic characteristics or functional convenience alone.



Across the analytical stages, Visual Engagement emerged as the strongest determinant of attitudinal loyalty, followed by Trust. This indicates that visually compelling presentation, aesthetic coherence, and culturally meaningful storytelling play a critical role in fostering emotional attachment to traditional silver jewelry in digital environments. Trust-related perceptions further reinforce loyalty by enhancing confidence, authenticity, and credibility in online transactions.

Demographic factors such as age, occupation, and shopping frequency show limited direct influence on loyalty outcomes. Instead, customer heterogeneity is better explained through experiential and psychological profiles, as revealed by the segmentation analysis. This highlights the superiority of experience-based segmentation over traditional demographic targeting for artisanal brands in digital markets.

From a managerial perspective, the findings suggest that analytics-driven insights enable firms to prioritize the experience dimensions that truly shape loyalty. For brands such as Đậu Bạc, sustainable success in e-commerce lies not in price competition or convenience alone, but in transforming cultural heritage into a visually engaging, trustworthy, and emotionally resonant digital brand experience.

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APPENDICES

CODE AVAILABILITY:

The data analysis code used in this study is available on Google Colab and can be accessed via the following link: [[Google Colab link: insert link here](#)]

The code covers the full analytical pipeline required for the Data Analysis course, including data cleaning and screening (duplicate removal), descriptive statistics, reliability analysis (Cronbach's Alpha), exploratory factor analysis (EFA), customer segmentation using K-means clustering, independent samples t-test, one-way ANOVA, correlation analysis, multicollinearity diagnostics (VIF and Condition Index), and multiple linear regression analysis.

ACADEMIC DISCUSSION AND METHODOLOGICAL EXCHANGE:

Throughout the research process, the research team actively engaged in academic discussions and methodological exchanges with other student groups in the course (Groups 5, 6, and 7). These exchanges focused on selecting appropriate data analysis techniques, interpreting statistical outputs, validating assumptions, and ensuring consistency between theoretical constructs and empirical testing. Such discussions contributed to improving methodological rigor, analytical accuracy, and alignment with course requirements.

AI USAGE STATEMENT

AI tools were used in a limited, transparent, and responsible manner to support selected stages of this study, in line with the academic integrity requirements of the Marketing Data Analysis course. ChatGPT was utilized to assist with idea clarification, academic language refinement, and structural improvement of written sections, particularly to enhance clarity, coherence, and consistency in reporting analytical results and managerial implications.

Additionally, AI support was applied to summarize theoretical concepts from assigned academic readings and to refine research descriptions without altering analytical content. Google Gemini was used exclusively for generating illustrative images for visualization purposes in the report and presentation slides, based on descriptive prompts only.